Request

How much water is stored in the upper 5 m at this montane catchment site, and how did it change between the spring and late-summer surveys?

Retrieval (RAG) — On

- Library docs and docstrings 2,604
- Curated catalog metadata 512
- Published studies 296

Skills

Procedures the agent reads when a task calls for them

Methods

- ✓ ERT
- Seismic
- GPR
- ✓ EM
- Borehole

Shared

- Data quality
- Inversion settings
- ✓ Time-lapse
- Interfaces
- ✓ Petrophysics
- ✓ Uncertainty

Agent trace

Illustrative replay with sample values · nothing is computed — Replay

Understand the request — Mid-tier · 0.9 s

Site: **a montane catchment site (sample)** · product **subsurface water storage** · electrical resistivity and electromagnetics.

Ground the request — 7 passages · 3 sources

Search over 3,412 chunks; the top 7 are kept.

> *"Convert resistivity to water content using Waxman-Smits model."*

`PyHydroGeophysX/petrophysics/resistivity_models.py`

Load skills — 5 of 5 needed

Plan with structured output — Mid-tier · 1.4 s

Constrained decoding against plan schema v1.2; the plan validated on the first pass. 6 tool calls, 2 inferred values, each cited.

Call registered tools — 6 calls

`load_ert_surveys` → `invert_time_lapse` → `load_tdem_data` → `invert_tdem` → `convert_water_content` → `write_report`

Execute deterministically — Pinned environment

The validated plan runs in a pinned environment with no model in the loop, and writes the report, figures and data files with a provenance record.

Deliver — Open the result in the workspace →

Sample values

Storage, 0 to 5 m: **1,056 ± 38** mm of water, line mean

Seasonal change: **−110 ± 13** mm, between the spring and late-summer surveys

✓ Curator review required

After curator approval, deposit to HydroShare under CC BY 4.0, inherited from the source surveys.

Design preview. Values are illustrative and no computation is performed.